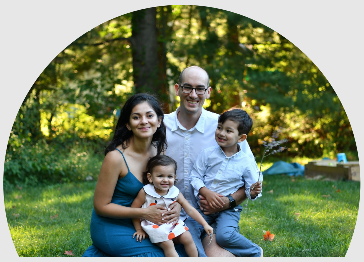

KirkMacKinnon

Ph.D. Candidate

July 2021

Molecular & Cellular Biology Program, UMASS Amherst

kirkmackinnon.github.io

kmackinnon@umass.edu

kirkmackinnon

About me

I am a molecular & computational biologist trained in the preparation and analysis of RNA-seq, ChIP-seq, DAP-seq, ATAC-seq, and GWA datasets. I am passionate about learning and staying up to date with the literature. I have a proven performance history of competent, publication worthy analyses in several different sequencing related disciplines.

Education

2011-2015 Bachelor of Science in Biology — UMASS Amherst
Amherst, MA

Research Experience

2015-Current Investigating the transcriptional regulation of growth in the model grass Brachypodium distachyon. — Hazen Lab
Ph.D candidate

2013-2015 Conducted time-lapse image analysis of Brachypodium distachyon in response to altered abiotic conditions. — Hazen Lab
Research Assistant

Teaching

2018-2020 Introductory Biology 1 (BIO 151). Graduate Teaching Assistant

Awards

2021	Howard E. and Margaret E. Barr Bigelow Memorial Scholarship.
2021	Lotta M. Crabtree Fellowship in Production Agriculture.
2020	Center for Agriculture, Food and the Environment Fellowship.
2019	Lotta M. Crabtree Fellowship in Production Agriculture.
2018	Center for Agriculture, Food and the Environment Fellowship.
2015	Outstanding Biology Research Award.
2013-2015	UMass Amherst Dean's List.

Publications

2021 Touch-triggered bZIP translocation regulates elongation and secondary wall biosynthesis
Joshua H. Coomey, Kirk J.-M. MacKinnon, Pubudu P. Handakumbura, Ian W. McCahill, Gina M. Trabucco, Jessica Mazzola, Nicole A. Leblanc, Rithany Kheam, Miriam Hernandez-Romero, Kerrie Barry, Lifeng Liu, Ji E. Lee, John P. Vogel, Ronan C. O'Malley, James J. Chambers, Samuel P. Hazen
- https://doi.org/10.1101/2021.02.03.429573

2020 Changes in ambient temperature are the prevailing cue in determining Brachypodium distachyon diurnal gene regulation — New Phytologist
Kirk J-M. MacKinnon, Benjamin J. Cole, Chang Yu, Joshua H. Coomey, Nolan T. Hartwick, Marie-Stanislas Remigereau, Tomás Duffy, Todd P. Michael, Steve A. Kay, Samuel P. Hazen
- https://doi.org/10.1111/nph.16507

2018 SECONDARY WALL ASSOCIATED MYB 1 is a positive regulator of secondary cell wall thickening in Brachypodium distachyon and is not found in the Brassicaceae — The Plant Journal
Pubudu P. Handakumbura, Kathryn Brow, Ian P. Whitney, Kangmei Zhao, Karen A. Sanguinet, Scott J. Lee, Jennifer Olins, Sandra P. Romero-Gamboa, Michael J. Harrington, Carlisle J. Bascom, Kirk J-M. MacKinnon, Michael T. Veling, Lifeng Liu, Ji E. Lee, John P. Vogel, Ronan C. O'Malley, Magdalena Bezanilla, Laura E. Bartley, Samuel P. Hazen
- https://doi.org/10.1111/tpj.14047

2018 Secondary Wall Regulating NACs Differentially Bind at the Promoter at a CELLULOSE SYNTHASE A4 Cis-eQTL — Frontiers in Plant Science
Jennifer R. Olins, Li Lin, Scott J. Lee, Gina M. Trabucco, Kirk J.-M. MacKinnon, Samuel P. Hazen
- https://doi.org/10.3389/fpls.2018.01895

2014 Daily Changes in Temperature, Not the Circadian Clock, Regulate Growth Rate in Brachypodium distachyon — PLoS ONE
Dominick A. Matos, Benjamin J. Cole, Ian P. Whitney, Kirk J.-M. MacKinnon, Steve A. Kay, Samuel P. Hazen
- https://doi.org/10.1371/journal.pone.0100072

Conference Presentations

Talks

2018 Exploring Diurnal Rhythmic Gene Expression in Brachypodium distachyon. ASPB Northeastern Section Meeting
Kirk J-M. MacKinnon, Benjamin J. Cole, Chang Yu, Marie Remigereau, Tomás Duffy, Steve A. Kay and Samuel P. Hazen.

Posters

2019 Changes in ambient temperature are the prevailing cue in determining Brachypodium distachyon diurnal gene regulation. Annual Symposium in Plant Biology
Kirk J-M. MacKinnon, Benjamin J. Cole, Chang Yu, Joshua H. Coomey, Nolan T. Hartwick, Marie-Stanislas Remigereau, Tomás Duffy, Todd P. Michael, Steve A. Kay, Samuel P. Hazen.

2019 Exploring Diurnal Rhythmic Gene Expression in Brachypodium distachyon. PAG XXVII
Kirk J-M. MacKinnon, Benjamin J. Cole, Chang Yu, Marie Remigereau, Tomás Duffy, Steve A. Kay and Samuel P. Hazen.

2014 Daily Changes in Temperature, Not the Circadian Clock, Regulate Growth Rate in Brachypodium distachyon UMass Undergraduate Research Symposium
Dominick A. Matos, Benjamin J. Cole, Ian P. Whitney, Kirk J.-M. MacKinnon, Steve A. Kay, Samuel P. Hazen.

Mentoring & Outreach

Train and mentor undergraduate assistants in the Hazen Lab

2020-Current Sanyam Shilpakar.
2017-2019 Chang Yu.
2018-2019 Alden Nichols.
2018-2018 Jacob Scott.
2017-2017 Crysta Daniels.

Creation of online tools for data exploration

Timecourse RNA-seq
https://hazenlab.shinyapps.io/shinyapp/

Touch Responsive Timecourse RNA-seq
https://hazenlab.shinyapps.io/swiztc/

Gain and loss of function RNA-seq
https://hazenlab.shinyapps.io/snd2r/